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Human Activity Recognition from Thigh and Wrist Accelerometry

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A. Objective

Shed some light on whether **the use of two IMUs enhances classification performance** for human activity recognition in comparison to a single sensor.

B. Motivation

Determine whether using more than one IMU will indeed be the best solution to implement going forward in the **IMPACT population cohort** (~200,000 individuals), which uses IMUs located on the thigh, wrist, and hip.

D. Results and Conclusions

- ✓ Accuracy increases by 15% when **combining data from the thigh and wrist IMUs** compared to using a single IMU.
- ❖ We propose **developing superclasses** as a potential strategy to further enhance the classification performance observed in this study.

C. Methods

- ❑ For this study, **public datasets** are used.
- ❑ **Convolutional Neural Networks** have been employed for training the classification models.

